MAT01B1 Final Exam Revision Pack

Prepared 21 October 2026, the night before the final. Built from your 2022 Exam C paper, the Exam A handoff and the ST2 handoff.

Where you stand

Exam C 2022 came out at about **45 out of 70 (64%)**, marked against my own solutions, so treat it as provisional by a few marks either way. Exam A was about 47 out of 70 (67%).

You lost 25 marks, and almost all of them were a slip of one line, not a missing topic. You got full marks on Questions 9 and 10, 11 of 12 on integration (Question 2) and 5 of 6 on differential equations. Your integration by parts, trig substitution, partial fractions, improper integral, matrices and binomial work were all sound.

Three things repeat across Exam A, Exam C and ST2, and they are where tomorrow's marks are:

1. Volumes of revolution: you lost marks on them in all three papers.
2. Theorem statements and proofs that are missing one required sentence.
3. A setup slip early in a question that then carries through the whole question, such as the domain in Question 4, which cost 4 marks by itself.

Marks lost in Exam C, by topic

1. Volumes of revolution, 5 marks (Question 5).
2. Parametric and polar, 5 marks (Question 7).
3. Domain, factorising and curve sketching, 4 marks (Question 4).
4. Surface area and arc length, 3 marks (Question 6a and 6c).
5. Theorem statements and the exactly one root argument, 3 marks (Question 3).
6. Arithmetic and multiple choice slips, 3 marks (1.3, 1.4, 1.5).
7. Finishing steps and one power rule slip, 2 marks (2c and 8b).

One thing I could not read with full confidence: the top left entry of your AC matrix in 9(b) has a stray mark on it. I read it as 6, which fits your trace of 16 and is correct. If it says 36 on your page, that would be a mark lost.

Exam C marked question by question

Question 1, 7 out of 10. You were right on 1.1, 1.2, 1.6, 1.7, 1.8, 1.9 and 1.10.

1. 1.3: the answer is (d) $\frac{5}{6}$. You chose (e). Your setup was right, the slip was combining the fractions. Use a common denominator of 6 and check with decimals.

2. 1.4: the answer is (b) 5. The average value is $\frac{5}{6}$, so six times it is 5. You chose (e) because the 1.3 slip carried over.
3. 1.5: the answer is (c) 27. A transpose swaps rows and columns, so b_{12} is the entry that was in row 2, column 1 of A. You chose (b) 10, which is the entry of A itself.

Question 2, 11 out of 12.

1. 2(a), 3 out of 3. The substitution and integration by parts were perfect. Simplify to $4e^3 - 2e^2$.
2. 2(b), 4 out of 4. Completing the square and the secant substitution were perfect.
3. 2(c), 2 out of 3. Partial fractions $A = 3$ and $B = -2$ were right, and so was the antiderivative. You did not finish: the answer is $\ln 18$, a finite number. The integral is not divergent because neither x nor $x - 4$ is zero anywhere on $[1, 2]$.
4. 2(d), 2 out of 2. Divergent, with the limit written. This is the finishing step you lost in Exam A, and you got it this time.

Question 3, 2 out of 5.

1. 3(a), 1 out of 2. The statement must say there exists $c \in (a, b)$ such that $f'(c) = 0$. You wrote only $f'(c) = 0$.
2. 3(b), 1 out of 3. You found $f'(x) = 3x^2 + 1 > 0$ and the sign change between 1 and 2, but called it Rolle's Theorem. Existence of a root is the Intermediate Value Theorem. Rolle's Theorem is used to rule out a second root. Full argument in Fix 4.

Question 4, 3 out of 7. Everything here traces to one slip: you wrote the domain as $-2 \leq x \leq 2$, from treating $2 - x^2$ as $(2 - x)(2 + x)$.

1. 4(a), 0 out of 1. The intercepts are $x = 0$ and $x = \pm\sqrt{2}$. You gave 0 and 2.
2. 4(b), 1 out of 2. Your second attempt for f' was correct, and so was the sign table with the local minimum at $x = -1$ and local maximum at $x = 1$. Missing: the values $f(-1) = -1$ and $f(1) = 1$, and the intervals ending at $\sqrt{2}$.
3. 4(c), 1 out of 2. Your numerator $2x^3 - 6x$ was correct. Then you factorised it as $2x(x^3 - 3)$. It is $2x(x^2 - 3)$.
4. 4(d), 1 out of 2. The shape was right: minimum on the left, maximum on the right, through the origin. The end points should be at $\pm\sqrt{2}$, and labelled.

Question 5, 4 out of 9.

1. 5(a), 1 out of 4. You used shells with radius $2 - x$. The axis $y = 2$ is horizontal and a vertical strip is perpendicular to it, so this is washers. Worked in Fix 1.
2. 5(b)(i), 1.5 out of 2.5. Intersection $x = 0$ and $x = 2$ was right and the shell setup was right. You did not evaluate the integral.
3. 5(b)(ii), 1.5 out of 2.5. Your integrand $y - \frac{y^4}{64}$ was exactly right. The limits should be 0 to 4, not 0 to 2, and it was not evaluated. Both methods give $\frac{24\pi}{5}$.

Question 6, 4 out of 7.

1. 6(a), 2 out of 3. The answer $\ln(2+\sqrt{3})$ is right. But you wrote the integral of $\sec^2 x$, then integrated $\sec x$. The square root of $\sec^2 x$ is $\sec x$, so write that line correctly.
2. 6(b), 1 out of 1. You named continuity on $[1,2]$ and the Mean Value Theorem for integrals. Add the conclusion sentence next time.
3. 6(c), 1 out of 3. Formula and derivative were right. The y factor cancels against the denominator, which leaves a constant. You kept a stray 1 over the square root. The sphere needs limits $-r$ to r, and the answer is $4\pi r^2$.

Question 7, 1 out of 6.

1. 7(a), 1 out of 2. You solved $y = 2t - 1$ as $t = \frac{y-1}{2}$. It is $t = \frac{y+1}{2}$. Your sketch had the right kind of shape and an arrow.
2. 7(b), 0 out of 2. The answer is $2xy = 1$. Worked in Fix 3.
3. 7(c), 0 out of 2. You were about to differentiate a fraction twice. Cancel the t first and it is short. Answer $(1, 0)$. Worked in Fix 3.

Question 8, 5 out of 6.

1. 8(a), 3 out of 3. Integrating factor, product rule step and $y = \frac{-\cos x + C}{x^2}$ were all correct.
2. 8(b), 2 out of 3. Separation and the right hand integral were right. The slip: the integral of $y^{-1/2}$ is $2y^{1/2}$, not $\frac{1}{2}y^{1/2}$. The answer is $y = (\ln x)^4$.

Question 9, 6 out of 6. AB is undefined because the inner sizes 3 and 2 do not match. $AC = \begin{bmatrix} 6 & 9 \\ 10 & 10 \end{bmatrix}$. The trace is 16. $B^T A$ is the 3×3 matrix you wrote.

Question 10, 2 out of 2. $r = 8$, coefficient $\binom{12}{8} 2^8 = 126\,720$.

Fix these first

Each fix shows the exact correction from your paper, then the rule that prevents it. They are in order of marks lost.

Fix 1: Volumes of revolution

Rule: draw the axis and one strip. A strip perpendicular to the axis gives washers. A strip parallel to the axis gives shells.

Question 5(a). Rotate the region between $y = \cos x$ and $y = \sin x$, for $0 \le x \le \frac{\pi}{4}$, about the line $y = 2$.

The axis $y = 2$ is horizontal. A vertical strip stands perpendicular to it, so this is washers with dx. The radii are distances from the line $y = 2$ down to each curve. At $x = 0$, $\sin x = 0$ and $\cos x = 1$, so $\sin x$ is farther from the axis: outer radius $2 - \sin x$, inner radius $2 - \cos x$.

$$V = \pi \int_0^{\pi/4} \left[(2 - \sin x)^2 - (2 - \cos x)^2 \right] dx$$

$$(2-\sin x)^2-(2-\cos x)^2=4(\cos x-\sin x)+\sin^2 x-\cos^2 x=4(\cos x-\sin x)-\cos 2x$$

$$V=\pi\Big[\,4(\sin x+\cos x)-\tfrac{1}{2}\sin 2x\,\Big]_0^{\pi/4}=\pi\Big[\big(4\sqrt{2}-\tfrac{1}{2}\big)-4\Big]=\pi\Big(4\sqrt{2}-\tfrac{9}{2}\Big)\approx 3.63$$

Check: $4\sqrt{2}\approx 5.657$, minus 4.5 is 1.157, which is positive, as a volume must be. Why shells failed: a shell radius is the distance from the strip to the axis. For a vertical strip and a horizontal axis that distance is 2 minus the height of the strip, not $2-x$.

Question 5(b). The curves $y=x^2$ and $y=2\sqrt{2}\sqrt{x}$ are rotated about the y axis. Setting them equal gives $x^4=8x$, so $x=0$ or $x=2$, and y runs from 0 to 4. At $x=1$ the root curve is 2.83 and the parabola is 1, so the root curve is on top.

Method (i), shells with x. The axis is vertical and a vertical strip is parallel to it.

$$V=2\pi\int_0^2 x\big(2\sqrt{2}\,\sqrt{x}-x^2\big)\,dx=2\pi\Big[\tfrac{4\sqrt{2}}{5}x^{5/2}-\tfrac{x^4}{4}\Big]_0^2=2\pi\Big(\tfrac{32}{5}-4\Big)=\tfrac{24\pi}{5}$$

Method (ii), washers with y. A horizontal strip is perpendicular to the vertical axis. Solve each curve for x: $y=x^2$ gives $x=\sqrt{y}$, and $y=2\sqrt{2}\sqrt{x}$ gives $x=\tfrac{y^2}{8}$. At $y=1$, $\sqrt{y}=1$ and $\tfrac{y^2}{8}=0.125$, so $\sqrt{y}$ is the outer radius.

$$V=\pi\int_0^4\Big[(\sqrt{y})^2-\Big(\tfrac{y^2}{8}\Big)^2\Big]dy=\pi\int_0^4\Big(y-\tfrac{y^4}{64}\Big)dy=\pi\Big[\tfrac{y^2}{2}-\tfrac{y^5}{320}\Big]_0^4=\pi\Big(8-\tfrac{16}{5}\Big)=\tfrac{24\pi}{5}$$

Both methods give $\tfrac{24\pi}{5}$. In the exam, if you have time, doing the second method is your best check.

Fix 2: Domain, factorising and the curve sketch

Rule: for a square root, the inside must be zero or more, and $2-x^2$ is not $4-x^2$. Practice on a different function: $g(x)=x\sqrt{3-x^2}$. We need $3-x^2\geq 0$, so $x^2\leq 3$, so $-\sqrt{3}\leq x\leq\sqrt{3}$. Test $x=2$: $3-4=-1$, which is not allowed, so $x=2$ is outside.

Question 4, $f(x)=x\sqrt{2-x^2}$. The domain is $-\sqrt{2}\leq x\leq\sqrt{2}$ (about ± 1.414). Intercepts: $f=0$ when $x=0$ or $2-x^2=0$, so $x=0$ and $x=\pm\sqrt{2}$.

$$f'(x)=\sqrt{2-x^2}-\frac{x^2}{\sqrt{2-x^2}}=\frac{2-2x^2}{\sqrt{2-x^2}}=\frac{2(1-x)(1+x)}{\sqrt{2-x^2}}$$

The denominator is positive on the domain, so the sign of f' is the sign of $(1-x)(1+x)$: negative for $x<-1$, positive between -1 and 1, negative for $x>1$. So f decreases on $(-\sqrt{2},-1)$ and $(1,\sqrt{2})$, and increases on $(-1,1)$. Local minimum $f(-1)=-1$. Local maximum $f(1)=1$.

$$f''(x)=\frac{2x\,(x^2-3)}{(2-x^2)^{3/2}}$$

On the domain $x^2\leq 2$, so x^2-3 is always negative. The sign of f'' is therefore opposite to the sign of x. Concave up on $(-\sqrt{2},0)$, concave down on $(0,\sqrt{2})$. The inflection point is $(0,0)$, because the sign changes there. Your $\sqrt[3]{3}$ is about 1.44, which is outside the domain, so it could never be an answer. A candidate outside the domain is always a sign of an earlier slip.

Sketch: start at $(-\sqrt{2}, 0)$ with a steep downward tangent, fall to the minimum $(-1, -1)$, rise through the inflection $(0, 0)$, reach the maximum $(1, 1)$, and fall steeply to $(\sqrt{2}, 0)$. The function is odd, $f(-x) = -f(x)$, so the picture is symmetric about the origin, which is a quick check.

Fix 3: Parametric and polar (Question 7)

Rule: after you solve for a letter, plug in one point to check it. If $t = 1$ gives $y = 1$, then $t = \frac{y+1}{2}$ must give 1 when $y = 1$. Yours, $\frac{y-1}{2}$, gives 0, so the check catches it.

7(a). $x = t^2 + t$, $y = 2t - 1$, $-1 \le t \le 1$. Points: $t = -1$ gives $(0, -3)$, $t = 0$ gives $(0, -1)$, $t = 1$ gives $(2, 1)$. Eliminating t:

$$t = \frac{y+1}{2}, \qquad x = \left(\frac{y+1}{2}\right)^2 + \frac{y+1}{2} = \frac{(y+1)(y+3)}{4}$$

This is a parabola opening to the right, with its vertex at $y = -2$, $x = -\frac{1}{4}$. The arrow points up the curve, from $(0, -3)$ to $(2, 1)$, because y grows as t grows.

7(b). Polar equation $r^2 \sin 2\theta = 1$. Use $\sin 2\theta = 2 \sin\theta \cos\theta$, then $x = r\cos\theta$ and $y = r\sin\theta$.

$$r^2 \sin 2\theta = 2\,(r\cos\theta)(r\sin\theta) = 2xy = 1 \quad \Longrightarrow \quad xy = \tfrac{1}{2}$$

This is a hyperbola. Your notes name the parabola and the circle but not the hyperbola, so only the equation $2xy = 1$ is needed for the marks. Your line $x^2 + y^2 = 2xy$ would be true only when $x = y$, so it cannot be the whole curve.

7(c). $x = 1 - t^2$, $y = t^7 + t^5$. Cancel the t before differentiating a second time.

$$\frac{dy}{dx} = \frac{7t^6 + 5t^4}{-2t} = -\frac{7t^5 + 5t^3}{2}$$

$$\frac{d^2y}{dx^2} = \frac{\dfrac{d}{dt}\left(-\dfrac{7t^5 + 5t^3}{2}\right)}{-2t} = \frac{35t^4 + 15t^2}{4t} = \frac{5t\,(7t^2 + 3)}{4}$$

This is zero only at $t = 0$, because $7t^2 + 3$ is always positive. It changes from negative to positive there, so $t = 0$ is an inflection. The point is $x = 1$, $y = 0$, which is $(1, 0)$.

Fix 4: Theorem statements and the exactly one root argument (Question 3)

Rule: every theorem sentence has hypotheses and a conclusion, and each part is a mark.

Rolle's Theorem. If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

3(b), $x^3 + x - 3 = 0$ has exactly one root. Let $f(x) = x^3 + x - 3$. It is continuous and differentiable everywhere.

1. At least one root (Intermediate Value Theorem): $f(1) = -1 < 0$ and $f(2) = 7 > 0$, so there is a root in $(1, 2)$.
2. At most one root (Rolle's Theorem): suppose two roots $a < b$. Then $f(a) = f(b) = 0$, so Rolle gives $c \in (a, b)$ with $f'(c) = 0$.
3. But $f'(x) = 3x^2 + 1 \ge 1$ for every x, so $f'(c) = 0$ is impossible. Contradiction, so there is exactly one root.

This exact question is already in your own Proofs Final sheet, under Rolle's Theorem and exactly one root. Read it once more tomorrow.

6(b). Write: $f(x) = \ln x$ is continuous on $[1, 2]$, so by the Mean Value Theorem for integrals there exists $c \in [1, 2]$ such that $f(c) = \dfrac{1}{2-1}\displaystyle\int_1^2 \ln x \, dx$, which is the average value of f.

Fix 5: Arc length and surface area (Question 6)

Rule: write the square root of the squared line exactly, and keep the y factor in surface area, because it often cancels.

6(a). $y = \ln(\cos x)$, $0 \le x \le \frac{\pi}{3}$. Here $\frac{dy}{dx} = -\tan x$, and $1 + \tan^2 x = \sec^2 x$, so the root is $\sec x$, not $\sec^2 x$.

$$L = \int_0^{\pi/3} \sqrt{1 + \tan^2 x}\, dx = \int_0^{\pi/3} \sec x \, dx = \Big[\ln|\sec x + \tan x|\Big]_0^{\pi/3} = \ln(2 + \sqrt{3})$$

6(c), surface area of a sphere of radius r. Rotate $y = \sqrt{r^2 - x^2}$ about the x axis, $-r \le x \le r$. Then $\frac{dy}{dx} = \frac{-x}{\sqrt{r^2-x^2}}$, so the root becomes $\frac{r}{\sqrt{r^2-x^2}}$.

$$S = \int_{-r}^{r} 2\pi\sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}}\, dx = \int_{-r}^{r} 2\pi r \, dx = 2\pi r\,(2r) = 4\pi r^2$$

The y factor cancels with the denominator, which leaves a constant. No trig substitution is needed. If you ever need one, you have probably dropped the y factor.

Fix 6: Small slips that cost a mark each

Fractions (1.3 and 1.4). Use a common denominator of 6.

$$\int_1^2 (x^2 - x)\, dx = \Big[\tfrac{x^3}{3} - \tfrac{x^2}{2}\Big]_1^2 = \Big(\tfrac{8}{3} - 2\Big) - \Big(\tfrac{1}{3} - \tfrac{1}{2}\Big) = \tfrac{2}{3} + \tfrac{1}{6} = \tfrac{5}{6}$$

So 1.3 is (d), and the average value is $\frac{5}{6}$, so 1.4 is $6 \cdot \frac{5}{6} = 5$, option (b). Both answers were listed, so none of the above was never needed.

Transpose (1.5). The transpose swaps rows and columns, so entry b_{12} of the transpose is the entry a_{21} of the original. Here that is 27, option (c). Test with a small example: if A has rows $(1, 2)$ and $(3, 4)$, the transpose has rows $(1, 3)$ and $(2, 4)$, and its top right entry is 3, which was the bottom left of A.

Power rule (8b). Add one to the power and divide by the new power. For $y^{-1/2}$ the new power is $\frac{1}{2}$, and dividing by $\frac{1}{2}$ means multiplying by 2.

$$\frac{dy}{4\sqrt{y}} = \frac{\ln x}{x}\, dx \implies \tfrac{1}{4} \cdot 2\sqrt{y} = \tfrac{(\ln x)^2}{2} + C \implies \sqrt{y} = (\ln x)^2 + K$$

At $x = e$, $y = 1$: $1 = 1 + K$, so $K = 0$, and $y = (\ln x)^4$. Check by differentiating: $\frac{dy}{dx} = \frac{4(\ln x)^3}{x}$, and the right side $\frac{4\sqrt{y}\,\ln x}{x}$ with $\sqrt{y} = (\ln x)^2$ gives the same.

Finish the answer (2c). After the antiderivative, evaluate and simplify.

$$3\ln 2 - 2(\ln 2 - \ln 3) = \ln 2 + 2\ln 3 = \ln 18$$

If an integral has no point where the bottom is zero on the interval, it is not improper, so "divergent" is not an option.

What you are supposed to know

This list follows what recurs in the past papers (Exams A, B and C, the 2023 papers and the semester tests). Your own Summary Final and Proofs Final notes hold the full versions.

Statements, word for word

Every hypothesis and the conclusion is a mark.

1. **Fermat's Theorem:** if f has a local maximum or minimum at c and $f'(c)$ exists, then $f'(c) = 0$.
2. **Rolle's Theorem:** if f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.
3. **Mean Value Theorem:** if f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.
4. **Mean Value Theorem for integrals:** if f is continuous on $[a, b]$, then there exists $c \in [a, b]$ such that $f(c) = f_{\text{ave}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$. Then solve $f(c) = f_{\text{ave}}$ for c.
5. **Intermediate Value Theorem:** if f is continuous on $[a, b]$ and N is between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ with $f(c) = N$.
6. **Extreme Value Theorem:** a continuous function on a closed interval attains a maximum and a minimum.

Proofs your notes rank as most likely: the Mean Value Theorem via Rolle, the Mean Value Theorem for integrals (two proofs), Fermat's Theorem, Rolle with exactly one root, the polar slope derivation, the binomial sigma form and the circumference as an improper integral. For Fermat, start from $f(c + h) \le f(c)$ for small h.

Formulas

$$\text{Area} = \int_a^b (\text{top} - \text{bottom})\,dx, \qquad f_{\text{ave}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$\text{Washers: } V = \pi \int (R^2 - r^2), \qquad \text{Shells: } V = 2\pi \int (\text{radius})(\text{height})$$

$$\text{Arc length: } L = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx, \qquad \text{Surface (about the } x \text{ axis): } S = \int 2\pi\, y\, ds$$

$$\text{Separable: } \int \frac{dy}{g(y)} = \int f(x)\,dx \qquad \text{Linear: } y' + Py = Q,\ \ I = e^{\int P\,dx},\ \ (Iy)' = IQ$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d^2y}{dx^2} = \frac{\dfrac{d}{dt}\left(dy/dx\right)}{dx/dt}$$

$$x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2, \qquad \frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k, \qquad \text{general term } \binom{n}{k} a^{n-k}b^k$$

Decision rules

1. **Washer or shell:** strip perpendicular to the axis is a washer, parallel to the axis is a shell. Horizontal axis with shells means dy.
2. **Square root of a squared line:** write the root before integrating ($\sqrt{\sec^2 x} = \sec x$).
3. **Trig substitution:** $\sqrt{a^2 - x^2}$ uses $x = a\sin\theta$, $\sqrt{a^2 + x^2}$ uses $x = a\tan\theta$, $\sqrt{x^2 - a^2}$ uses $x = a\sec\theta$. Complete the square first when you see $x^2 + 2x$.
4. **Partial fractions:** a linear factor gets a constant over it, a repeated factor gets one term per power, an irreducible quadratic gets $Ax + B$ over it. Check the degree on top first.
5. **Improper integral:** write the limit, evaluate, and say convergent or divergent.
6. **Curve sketch order:** domain, intercepts, f' and its sign table, f'' and its sign table, then the sketch. Check every critical point is inside the domain.
7. **Identities:** $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$, $1 + \sinh^2 x = \cosh^2 x$, $\sin 2\theta = 2\sin\theta\cos\theta$, $\cos 2\theta = \cos^2\theta - \sin^2\theta$.
8. **Multiple choice:** simplify your answer into the form of each option, or test one value, before choosing none of the above.

Matrices and linear systems

1. The product AB is defined only when the columns of A equal the rows of B, and the size is rows of A by columns of B.
2. The transpose swaps rows and columns. The trace is the sum of the diagonal entries and needs a square matrix.
3. Row echelon form: zero rows at the bottom and leading 1s stepping to the right. Reduced form also has zeros above and below each leading 1.
4. The last column of an augmented matrix is the right hand side, even when no line is drawn. A row of zeros with a nonzero number on the right means no solution.
5. A free variable is a column with no leading 1. Give each free variable its own parameter.
6. After solving, plug your answer into the original equations.

Plan for the trip and tomorrow morning

Do not start another full paper tonight. You have seen the same slips in three papers, so the best use of the time is to fix those, not to find new ones. About 90 minutes in total.

1. 25 minutes, volumes. Redo Question 5(a) on paper without looking, then 5(b) with both methods. Draw the axis and one strip first, every time.

2. 15 minutes, statements. Write Rolle, the Mean Value Theorem, the Mean Value Theorem for integrals, Fermat and the Intermediate Value Theorem from memory, then check them against this document. Write the exactly one root argument once.
3. 20 minutes, Question 4 and Question 7 from a blank page. Domain first. For 7(c), cancel t before the second derivative.
4. 15 minutes, multiple choice routine. Redo 1.3, 1.4 and 1.5 in your head, then say out loud what you check before choosing none of the above.
5. 15 minutes, read the formulas section once, then close it.

Before you sleep, read only the first page of this document, where your three repeating slips are listed.

In the exam room

1. Your plan from your own notes: set a timer for 30 minutes left if a phone is allowed, and check the rule first.
2. About one minute per mark. Circle anything that stalls you and move on. You were 10 minutes over on this paper, so skipping early is worth more than finishing one question.
3. If you blank, write the first line of the theorem or the setup. Setup lines earn marks even when the finish slips.
4. Before each volume question, draw the axis and one strip, and say washer or shell.
5. At the end, plug in or differentiate to check, and check every multiple choice option against your own answer.

You finished a full 70 mark paper with correct work on integration, matrices, the binomial question, differential equations and your improper integral. The marks you lost are the kind that a calm recheck recovers.